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Dragonfly (Odonata) communities at three lotic sites with different hydrological characteristics

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Abstract

At the nature reserve “Bosco della Fontana” (Lombardy, Italy) the communities of larval and adult dragonflies were studied at three sites in small streams. The larval communities were investigated by collecting exuviae. Site 1, a stream section which falls dry in March–April due to human intervention, had a community of larval dragonflies where species typical of running water were almost absent. Here *Ischnura elegans* was the dominant species, accounting for more than 64% of the total number of exuviae. Additionally, other species typical for stagnant water were present. However, reproductively active adults of rheophilous species were dominant when flowing water was abundant at this site in summer. In contrast, at the two other sites, where flowing water is present throughout the entire year, rheophilous species were dominant in the larval community and in the adult communities. It is concluded that streams with a highly modified hydrology may represent “ecological traps” for specialized species and that dragonfly surveys solely based on the observation of adults can be misleading.

Keywords: *Odonata, exuvia, dragonfly community, streams, phenology, Italy*

Introduction

Many freshwater systems in Europe and worldwide have been profoundly changed by man, often for agricultural purposes (Revenge & Kura 2003). In some cases they present abiotic (and biotic) conditions normally not found in nature. These altered hydrological systems (e.g. ditches) constitute novel habitats for aquatic species.

In Europe, dragonflies (Odonata) are a well studied insect group (e.g. Askew 1988) and are often used as bioindicators (Chovanec 1994; Sahlén & Ekkestubbe 2001; Briers & Biggs 2003). Additionally, larval populations can easily be studied by collecting exuviae, a reliable indirect method of monitoring dragonfly populations (Foster & Soluk 2004). Corbet (1999) writes that “the prospect of being able to collect all exuviae [...] emerging from a habitat offers a powerful method for quantifying [...] emergence”. In most studies where exuviae are searched for it is not attempted to compare Odonata communities but the focus is on particular species, most commonly on Anisoptera (e.g. Wildermuth &

Knapp 1993; Jakob & Suhling 1999; Foster & Soluk 2004). Many other researchers have investigated only a specific period, covering less than the emergence period of the entire community (e.g. Tomc 1997; Bernier & Guilloux 2004). Studying the emergence of Odonata between early spring to late autumn is probably the best method to study the larval communities of Odonata, as the larvae of some species are very difficult to collect (e.g. larvae of *Somatochlora metallica*, personal observation). Moreover, larvae of different species live in different microhabitats and it is difficult not to bias in favour of “more accessible” species (e.g. Watson et al. 1982; Castella 1987).

The importance of habitat characteristics and local microhabitats of stream systems for the local dragonfly fauna has long been recognized (e.g. Aguesse 1960; Ferreras Romero 1988). Therefore, Odonata were chosen for an investigation into the differences between communities in three lotic habitats with different characteristics (e.g. Schridde & Suhling 1994).

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It is known that adult dragonflies use “proximate factors” such as vegetation or “polarization patterns” as a signal to identify suitable habitats (Buchwald 1992; Wildermuth & Horváth 2005), in order to maximise fitness. Probably, it is never a single factor, but always a combination of visual characteristics which leads the animals to a given habitat (Wildermuth 1994). Water bodies altered by man may either not provide the necessary proximate factors or exhibit the wrong ones (see also Hofmann & Mason 2005a). Thus, species might be excluded from habitats or fall victim to “ecological traps” (Schlaepfer et al. 2002; Robertson & Hutto 2006), recognizing suitable habitats by means of proximate factors when in reality habitat conditions are not suitable. Differences between the community of adults showing reproductive behaviour and the community of larvae completing development (as revealed by exuviae), might, particularly in highly altered environments, provide some insight into the proximate factors used by adult Odonata. Unfortunately, comparisons between adults versus larvae/exuviae are few (e.g. Amico et al. 2004).

For dragonflies, emergence is a critical period, the timing of which depends on many factors (e.g. Brodin & Johansson 2002; Corbet 2003). This leads to emergence periods differing substantially even for the same species within the same water system. Marked differences in the timing of emergence have been found between different years in *Cordulia aenea* (Linnaeus, 1758) (e.g. Wildermuth 1998) and in different sections with different thermal characteristics in *Chalcolestes viridis* (Van der Linden, 1825) (Westermann 2002). Also, the synchrony of emergence can differ markedly. Suhling and Müller (1996) provide an example for *Onychogomphus uncatus* (Charpentier, 1840), where in one population EM_{50} (the time elapsed between the first emergence and the median) was reached in less than 5 days, whereas in a nearby population EM_{50} was only reached after 36 days. The authors attributed this marked difference to the different temperature regimes of the two rivers.

The aim of the present study was fourfold:

1. To compare the larval communities of dragonflies of three lotic habitats with different characteristics so as to elucidate habitat requirements of the species present at Bosco della Fontana. The hypothesis was that species typical for streams should be less present in the site where the water flow is annually interrupted from the end of March to the end of April.

2. To compare the adult communities of dragonflies between the three sites. The hypothesis was that the adult communities should be similar between all sites, because during summer all sites should “appear” as suitable habitats (“proximate factors”) to adult rheophilous dragonflies.
3. To compare the similarity of the species composition of larvae and adult Odonata for each site. The hypothesis was that the site where the water flow is interrupted annually from the end of March to the end of April, should have clearly different larval and adult communities, whereas the species composition should be similar in sites where the water flow is continuous.
4. To compare the phenology of the emergence of the Odonata in the three sites. No a priori hypothesis was formulated.

In order to investigate the above aims it was decided to collect exuviae of all species throughout the entire emergence period, from early spring to late autumn. Only the collection of exuviae allows for comparison of emergence patterns.

Materials and Methods

The study area

The nature reserve “Bosco della Fontana” (coordinates 45°12' N, 10°44' E) is located near Marmirolo, about 5 km NW of Mantua, Lombardy, Italy (see Figure 1). The reserve covers an area of about 233 ha, 198 of which are occupied by woodland, 33 by meadows and 2 by an artificial wetland. The forest has had continuous woodland cover since at least 1600 AD. The landscape surrounding the reserve is highly modified by human activities (agriculture, rural settlements, industry) and thus the reserve is isolated from other woodlands (for further details see Mason 2002). The water system of Bosco della Fontana does not follow a natural cycle but is influenced by the water management of the surrounding agricultural areas.

The sites

Sites 1 and 2 (see Figure 1) were situated in the riverbed of the Rio Begotta: this small stream, which measures approximately 5–9 m in width, enters the reserve on its north-western side. One branch, which can be closed with a barrage, feeds an artificial wetland. The Rio Begotta flows via a ditch surrounding an ancient building (“Palazzina”) and

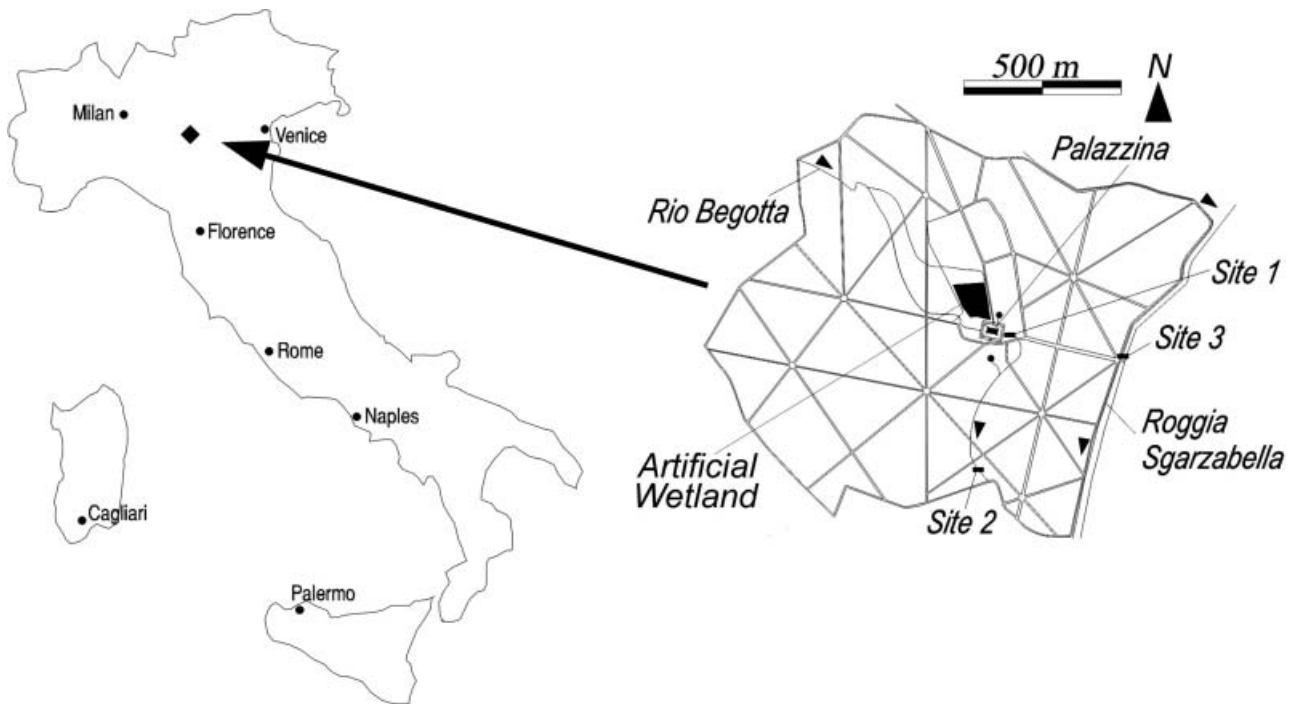


Figure 1. The location of Bosco della Fontana and of the three sites investigated. Arrows next to the streams indicate the direction of flow.

then through a section enclosed between two brick walls, which is approximately 80 m long and 5.5 m wide. The stream continues for approximately 450 m through the dense forest and exits the reserve on the south-western side. The southern section of the Rio Begotta has a more or less natural morphology and is supplied by a number of small springs. These, together with a small well, normally provide a continuous water supply. In contrast, parts of the northern sections normally fall dry during winter-spring.

Site 1 (see Figure 1) was situated in the section enclosed by brick walls, which measured 0.88 cm in height at this point. No trees grow near this stretch of the stream, which is thus open to the sun.

Site 2 (see Figure 1) was situated approximately 40 m before the Rio Begotta reaches the forest margin. Here, the stream measures 9.5 m in width. Due to the river morphology and to some recently fallen trees, this site was relatively open and received direct sunlight during summer.

In both Sites 1 and 2, a 16 m walkway (width 20 cm) was placed on 24 March 2004 in order to minimize disturbance while searching for exuviae, as suggested by Foster and Soluk (2004). The walkway consisted of 4 planks measuring 4 m × 0.2 m each, fixed with nails on 5 trunk sections (*Quercus rubra*, length 45–55 cm, diameter 35–40 cm). These walkways were placed in such a way as to cross the main microhabitats present (main channel, fine sediment,

dense vegetation, etc.). Exuviae were searched for on either side of the walkways within a distance of 50 cm, in an area equalling approximately 21 m².

Site 3 was situated on the eastern bank of the Roggia Sgarzabella: This small river arrives from the north and forms the eastern outer perimeter of the reserve. It measures approximately 4–5 m in width and has flowing water all year round. The river has a man-made profile. One result of human intervention is a reduction of habitat diversity, and in particular the elimination of pools and areas with shifting sediments. The banks of the Roggia Sgarzabella are steep and the morphology of the watercourse is relatively uniform. Moreover, the aquatic vegetation is regularly cut. Site 3 did not allow for the placement of a walkway. Here, the eastern bank was chosen to search for exuviae, between the water level and approximately 1 m above it, along a 10 m stretch.

In order to allow for quantitative collection of all exuviae (including Zygoptera) relatively small areas were selected compared to other studies.

Searching for exuviae and adults

The sites were searched once a week between 31 March 2004 and 4 November 2004 between 10 a.m. and 4 p.m. In 5 out of 32 cases the sampling of all sites was carried out in 2 days rather than on a single day. It was always attempted to find all exuviae

(including Zygoptera). Each exuvia was carefully removed from the substrate (plants, rocks, etc.) and placed in a collecting jar. All exuviae found in one day at one site were stored in a single jar. These were left open for 1–2 days indoors to allow the exuviae to dry out.

At each visit to the sites, the adult dragonflies present were also noted and their abundance in approximately 15 m of the stream estimated. Most individuals were identified “on the wing”. Whenever necessary a sweep-net was used to catch dragonflies for identification (e.g. *Sympetrum*).

The keys used for the determination of adult Odonata and exuviae were: Conci and Nielsen (1956), Carchini (1983), Askew (1988), Heidemann and Seidenbusch (1993), Suhling and Müller (1996), Kohl (1998), Lehmann and Nüß (1998).

Characteristics of the streams

Between 14 April 2004 and 05 January 2005, water depth was measured weekly at each site with a folding ruler. Conductivity and pH were measured at 5–8 cm below the surface with a “HANNA HI 98129” from 28 April 2004 to 04 November 2004. Temperature was measured with the same instrument at the following dates: winter (4 measurements: 22 December 2004, 20 January 2005, 02 February 2005, 02 March 2005), spring (9 weekly measurements: 21 April–16 June 2004), summer (13 weekly measurements: 23 June–16 September 2004) and autumn (7 weekly measurements: 22 September–4 November 2004). Vegetation (vegetation cover according to Reichelt & Wilmanns 1973), stream morphology and flow rate of the water were analysed on 23 August 2004.

Data analysis

In order to analyse differences in species frequencies the data for larvae and adults were compared using Pearson’s chi-squared test when the number of observations allowed for it.

In order to analyse the similarity of species composition “Chao’s estimator for the Jaccard Abundance-based index” (Chao et al. 2005) was employed (using EstimateS 8.0).

In order to analyse the emergence patterns the median and the “main emergence period” were employed, as suggested by Westermann (2002), rather than the commonly used value EM_{50} (the time elapsed between the first emergence and the median). The main reasons for this choice are (for more details see Westermann 2002):

1. the first exuvia is almost impossible to find. Therefore, the EM_{50} value is almost always based on larvae having emerged later and the EM_{50} value is thus almost always underestimated;
2. in a large population the likelihood of finding an early exuvia is greater. Thus, the EM_{50} value is related to population size.

In contrast, the median is largely independent against extreme values and is not related to population size. Moreover, medians can easily be compared statistically. In the present paper this was done using the Kruskal–Wallis test and the Mann–Whitney U Test (using OS4 Version 1, Revision 7 (© 2006)). As a measure of duration of emergence the period between the first and third quartile was calculated. This time span is called “main emergence period” (see Westermann 2002). This value comprises the timespan defined by the emergence of 25% of the individuals having emerged prior to the median and 25% of those emerged after the median. This value does not depend on the first or last exuvia collected and is therefore much more robust.

When multiple comparisons were carried out the sequential Bonferroni correction (Rice 1989) was applied.

Results

Between 14 April 2004 and 20 October 2004 a total of 575 exuviae belonging to 14 species were collected in the three sites. The numbers of exuviae for each species are given in Table I.

Table I. Numbers of exuviae collected at the three sites. “***” indicates species for which the number of observations allowed for Pearson’s chi-squared test to be carried out and for which the frequencies deviated highly significantly from being evenly distributed after sequential Bonferroni correction (Rice 1989).

	Site 1	Site 2	Site 3
<i>C. splendens</i> ***	3	30	92
<i>C. viridis</i>	0	0	3
<i>P. pennipes</i> ***	59	23	24
<i>P. nymphula</i>	0	4	0
<i>I. elegans</i> ***	164	0	17
<i>A. mixta</i>	4	0	0
<i>A. imperator</i>	2	0	0
<i>G. vulgatissimus</i>	0	2	2
<i>O. forcipatus</i> ***	0	15	50
<i>S. metallica</i> ***	0	5	19
<i>L. fulva</i> ***	23	3	0
<i>O. cancellatum</i>	0	0	1
<i>O. coerulescens</i> ***	0	14	13
<i>S. pedemontanum</i>	0	0	3

Table II. Similarities for species composition of larval and adult data using Chao's estimator for the Jaccard Abundance-based index (Chao et al. 2005).

	Site 1	Site 2	Site 3	Data adults
Site 1		0.998	0.973	
Site 2	0.424		0.989	
Site 3	0.711	0.91		
	Data larvae			

The frequencies of the exuviae data analysed deviated highly significantly from being evenly distributed in all cases (indicated with “***” in Table I) where the number of observations allowed for the chi-squared test to be carried out also after sequential Bonferroni correction (Rice 1989). The similarity indexes for the larval data (see Table II) show that the community of Site 1 was very different from Sites 2 and 3. The latter sites, in contrast, were similar, Chao's estimator being larger than 0.9.

A total of 2356 adult dragonflies belonging to 21 species were observed at the three sites. Table III gives the composition of the community of adult dragonflies for each of the three sites.

Table III. Numbers of adults observed at the three sites. “***” indicates species for which the number of observations allowed for Pearson's chi-squared test to be carried out and for which the frequencies deviated highly significantly from being evenly distributed after sequential Bonferroni correction (Rice 1989). “^” indicates species for which the number of observations allowed for Pearson's chi-squared test to be carried out and for which the data did not deviate from being evenly distributed after sequential Bonferroni correction (Rice 1989).

	Site 1	Site 2	Site 3
<i>C. splendens</i> ***	255	573	365
<i>P. pennipes</i> ^	106	84	102
<i>P. nymphula</i>	0	0	1
<i>I. elegans</i> ***	58	3	16
<i>C. tenellum</i>	0	0	1
<i>B. pratense</i>	0	4	0
<i>A. mixta</i>	13	0	3
<i>A. affinis</i>	1	1	0
<i>A. isosceles</i>	5	0	0
<i>A. imperator</i>	7	0	0
<i>A. parthenope</i>	1	0	0
<i>G. vulgatissimus</i>	0	1	0
<i>O. forcipatus</i> ^	8	5	5
<i>S. flavomaculata</i>	0	1	0
<i>S. metallica</i> ^	1	4	9
<i>L. fulva</i> ^	50	20	37
<i>O. cancellatum</i>	8	3	0
<i>O. coerulescens</i> ***	129	244	207
<i>S. depressiusculum</i>	14	1	0
<i>S. pedemontanum</i>	2	0	4
<i>S. striolatum</i>	2	1	1

The species most commonly observed were: *Calopteryx splendens* (Harris, 1782), *Orthetrum coerulescens* (Fabricius, 1798), *Platycnemis pennipes* (Pallas, 1771). After sequential Bonferroni correction (Rice 1989) the adult frequencies analysed (indicated with a “***” in Table III) deviated highly significantly from being evenly distributed for *C. splendens*, *Ischnura elegans* (Van der Linden, 1820) and *O. coerulescens*. The frequencies of *P. pennipes*, *Onychogomphus forcipatus* (Linnaeus, 1758), *Somatochlora metallica* and *Libellula fulva* Müller, 1764, on the other hand, did not differ significantly from being evenly distributed (indicated with a “^” in Table III). The similarity indexes for the adult data (see Table II) showed that the three communities were very similar with the Chao's estimator always exceeding 0.97.

When comparing the differences of the species composition of larvae and adult Odonata (see Table IV), Site 1 had the lowest similarity index of all three sites and thus larval and adult communities differed more when compared with Sites 2 and 3.

The phenology of emergence (as revealed by the exuviae) of species with a total of more than 50 exuviae collected (*C. splendens*, *I. elegans*, *P. pennipes*, *O. forcipatus*) is given in Figures 2–5. Data sets with $n \geq 15$ were compared statistically (*C. splendens*, *I. elegans*, *P. pennipes*, *O. forcipatus*).

Calopteryx splendens

Exuviae were found between 28 April 2004 and 21 October 2004 (Figure 2). As the sites were searched every week, it can be stated with confidence that the emergence of *C. splendens* lasted at least until mid October. At Site 2 the median was reached on 1 September 2004, whereas at Site 3 this value had already been reached on 19 August 2004, a highly significant difference (Mann–Whitney U test; $Z=3.45$; $P \leq 0.001$). The main emergence period lasted from 24 August 2004 to 9 September 2004 in Site 2 and from 25 July 2004 to 2 September 2004 in Site 3.

Table IV. Similarities for species composition between larval and adult communities using Chao's estimator for the Jaccard Abundance-based index (Chao et al. 2005).

	Similarity Index
Site 1	0.853
Site 2	0.977
Site 3	0.958

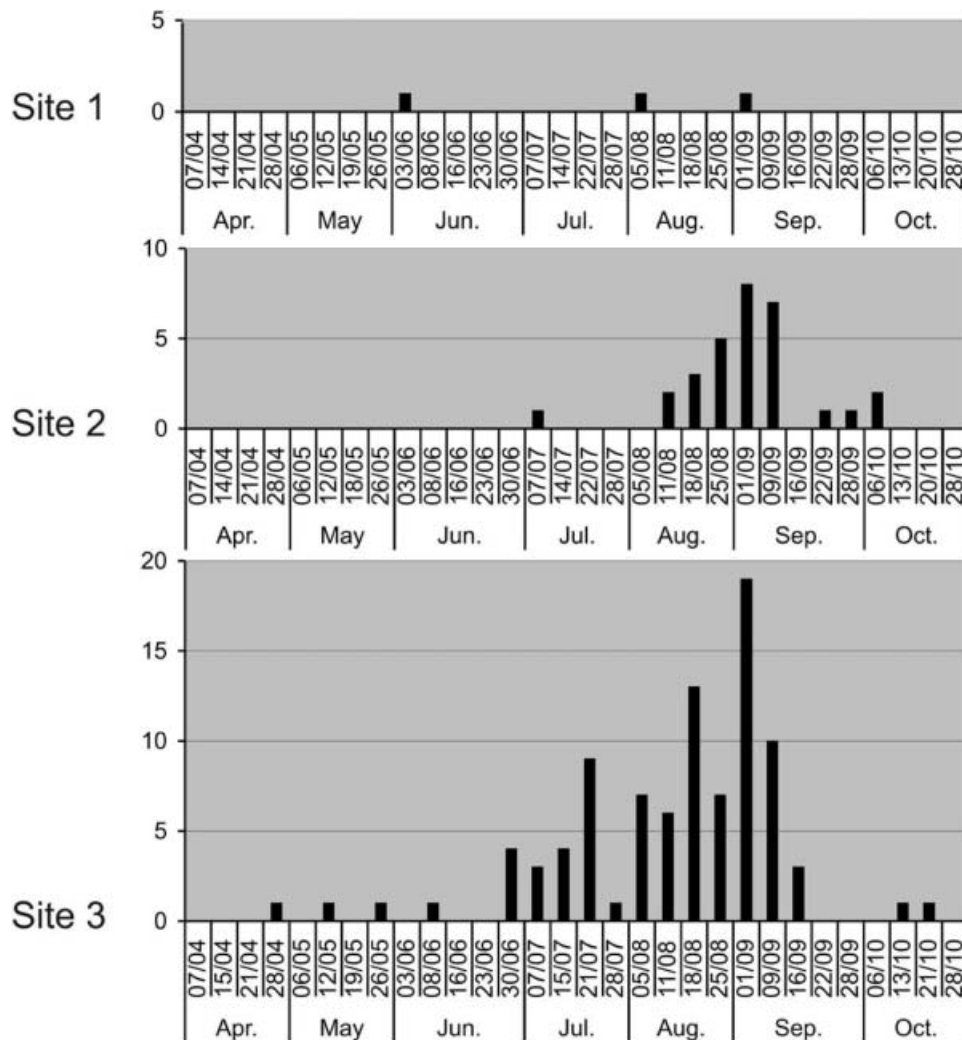


Figure 2. Emergence data of *Calopteryx splendens*.

Ischnura elegans

Exuviae were found between 21 April 2004 and 1 September 2004 (Figure 3). At Site 1 the median was reached on 08 June 2004, whereas at Site 3 this value was only reached on 7 July 2004, a highly significant difference (Mann–Whitney U test; $Z=4.136$; $P\leq 0.0001$). The main emergence period lasted from 30 May 2004 to 30 June 2004 at Site 1 and from 27 June 2004 to 16 July 2004 at Site 3.

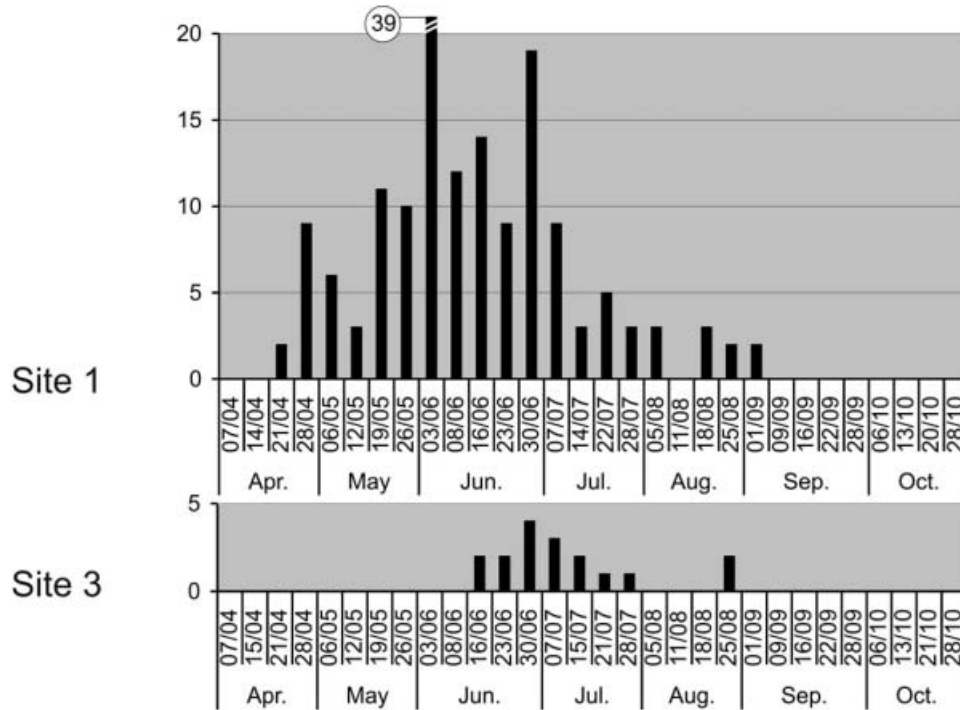
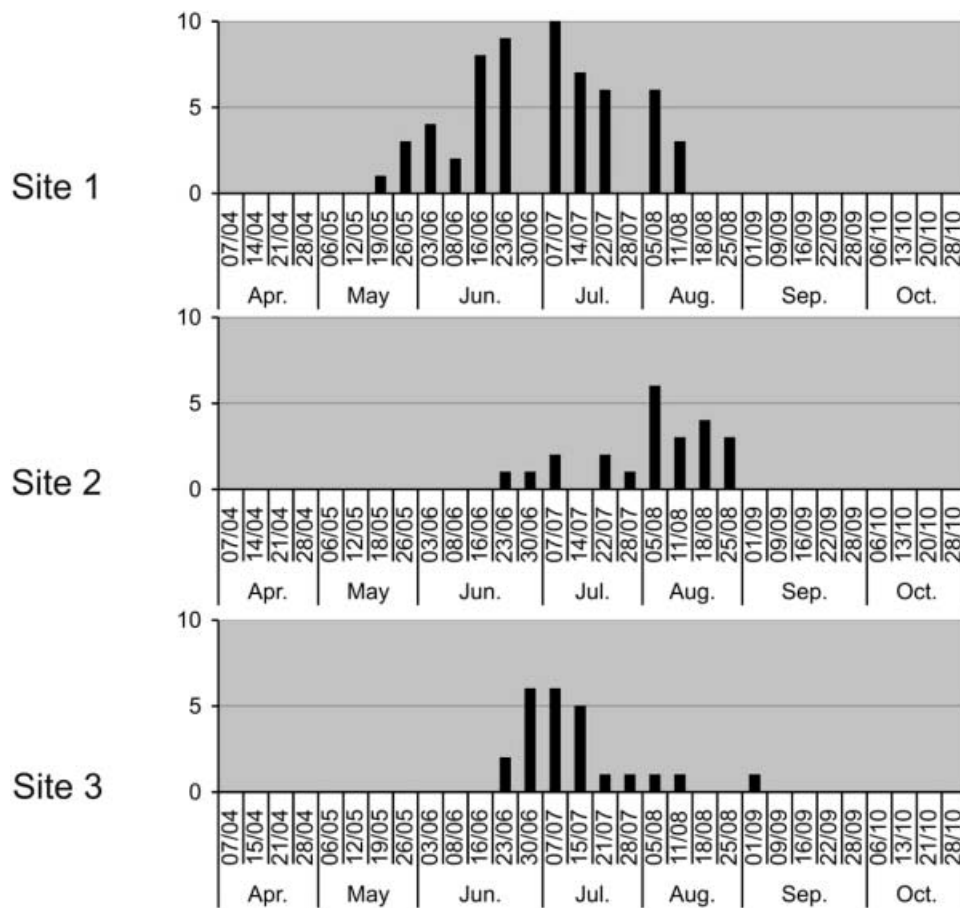
Platycnemis pennipes

Exuviae were found between 19 May 2004 and 1 September 2004 (Figure 4). At Sites 1 and 3 the median was reached on 5 July 2004 and 8 July 2004, respectively, whereas the median at Site 2 was reached only on 7 August 2004. This difference between sites was highly significant (Kruskal–Wallis H Test, $H_2=27.240$; $P\leq 0.0001$). The difference

between Site 2 and the other two sites was in both cases highly significant (Mann–Whitney U test; $Z>3.59$; $P\leq 0.001$), also after sequential Bonferroni correction (Rice 1989). At Bosco della Fontana the main emergence period lasted from 17 June 2004 to 18 July 2004 in Site 1, from 24 July 2004 to 17 August 2004 in Site 2 and from 1 July 2004 to 16 July 2004 in Site 3.

Onychogomphus forcipatus

Exuviae were found between 16 June 2004 and 22 September 2004 (Figure 5). At Site 2 the median was reached on 5 August 2004, whereas at Site 3 this value was reached on 22 July 2004, a highly significant difference (Mann–Whitney U test; $Z=2.390$; $P\leq 0.01$). The main emergence period lasted from 22 July 2004 to 10 September 2004 at Site 2 and from 5 July 2004 to 10 August 2004 at Site 3.

Figure 3. Emergence data of *Ischnura elegans*.Figure 4. Emergence data of *Platynemesis pennipes*.

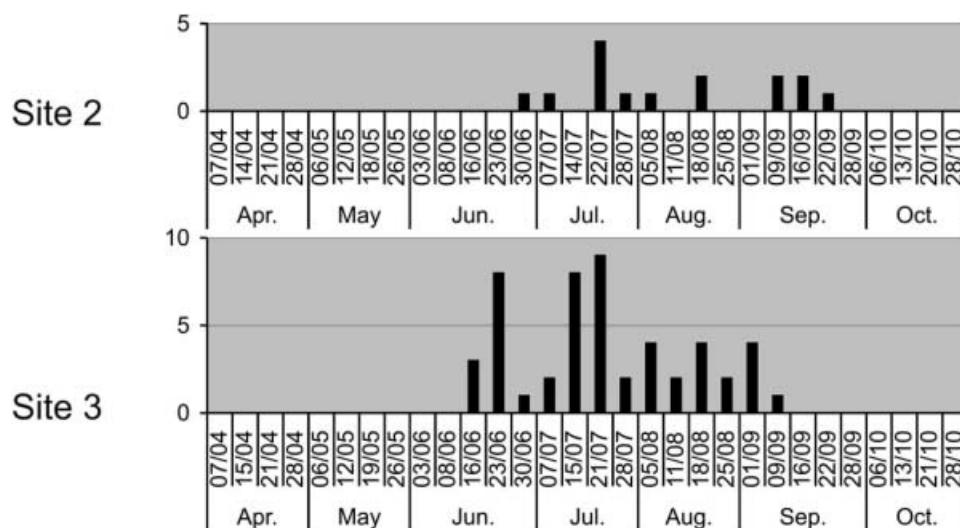


Figure 5. Emergence data of *Onychogomphus forcipatus*.

For *Somatochlora metallica*, exuviae were found between 15 July 2004 and 21 October 2004. The position of the last exuvia found suggested that it might have fallen there. The last exuviae, the emergence of which could reliably be dated, were found on 6 October 2004. Consequently, the main emergence period lasted from 6 August 2004 to 6 October 2004 at Site 3. For *Libellula fulva*, exuviae were found between 28 April 2004 and 7 July 2004 and the main emergence period lasted from 7 May 2004 to 28 May 2004 at Site 1. For *Orthetrum coerulescens*, exuviae were found between 16 June 2004 and 9 September 2004.

Fluctuations of water depth at the three sites are presented in Figure 6.

Conductivity and pH in all three sites varied only moderately. Their means with standard errors as well as the velocity of the water are given in Table V.

Vegetation and morphology of the sites are presented in Figures 7–9.

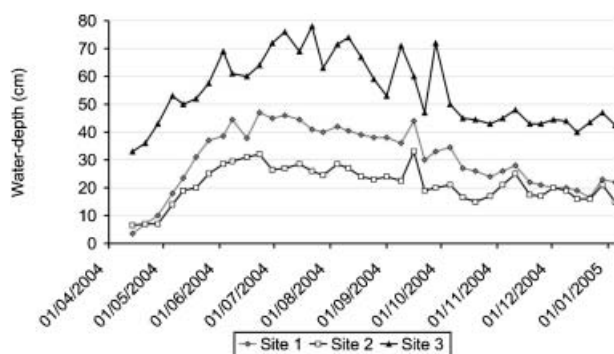


Figure 6. Water depth (cm) of the three sites investigated (weekly measurements).

In Table VI, temperatures are presented as averages.

Discussion

The larval communities (as revealed by exuviae)

At the three sites, exuviae of a total of 14 species were found. These correspond to 45% of the 31 Odonata species known from Bosco della Fontana (Hardersen 2004). The quantitative collection of all exuviae (including Zygoptera) in the relatively small areas resulted in 575 exuviae to be collected, a number which allowed meaningful comparisons.

At Bosco della Fontana, the larval communities from Site 1, the site where water flow was interrupted in March–April, differed substantially from those of the other two sites. The section of the Rio Begotta up-stream was very similar to Site 1 from a hydrological point of view. Therefore drift is highly unlikely to have resulted in the arrival of rheophilous species (e.g. *Calopteryx splendens*).

In order to analyse the differences from an ecological point of view Carchini's (1983) classification was used.

Table V. Characteristics of the water in the three sites investigated.

	Site 1	Site 2	Site 3
pH (standard error)	7.5 (0.2)	7.3 (0.2)	7.7 (0.2)
Conductivity (standard error)	398 (14)	411 (19)	406 (15)
Water flow (m s^{-1} , 23 August 2004)	0.3	0.7	0.8

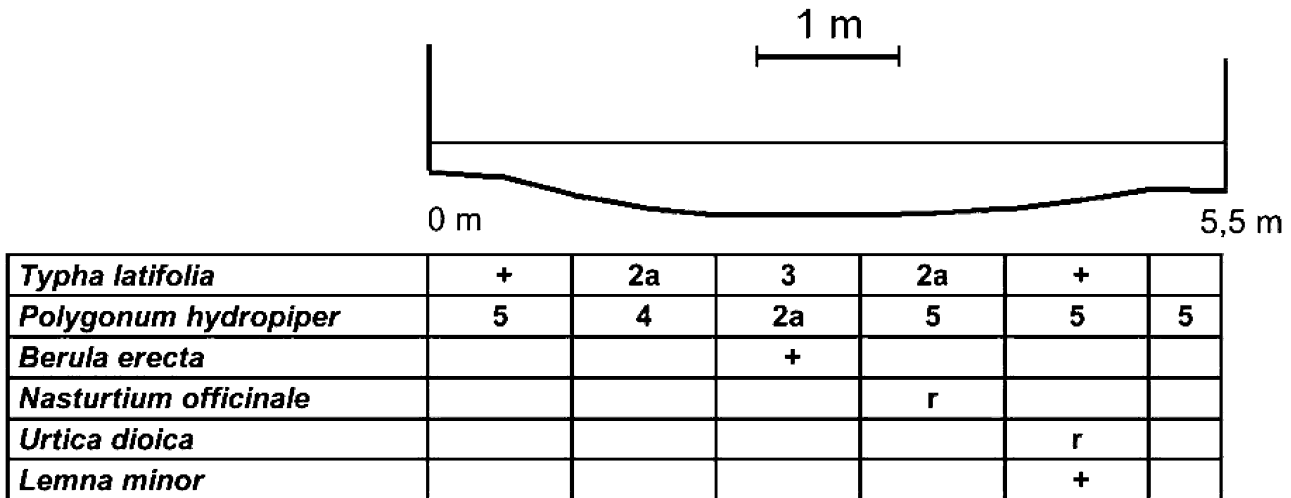


Figure 7. Site 1: morphology and vegetation (cover according to Reichelt & Wilmanns (1973); r: 1 individual, +: 2–5 individuals (<5% cover), 1: 6–50 individuals (<5% cover), 2m: >50 individuals (<5% cover), 2a: cover 5–15%, 2b: cover 16–25%, 3: cover 26–50%, 4: cover 51–75%, 5: cover 76–100%).

Site 1 was characterized by large fluctuations in water level and flow (Figure 6). From the end of March to the end of April the depth of water was only 3–5 cm at the point where measurements were taken (personal observation). Other parts of this section were dry and in the parts where some water was left no water flow was discernible. On 25 April 2004 the water level started to rise and a gentle water flow became obvious. From 03 May 2004 the water flow was substantial. Maximum flow was reached in June and July, with 47 cm of water depth, and declined continuously until early March 2005, when it reached similar levels to March 2004 (personal observation). With 0.3 m s^{-1} , water flow was slowest at Site 1 (Table V). Water was colder at Site 1 in comparison to the other sites during winter due to an even slower flow and reduced volume. At Site 1, plant species typical for flowing water were absent

and the vegetation was dominated by a species (*Typha latifolia*) which indicate fluctuating water levels with considerable time spans without water cover (Ellenberg et al. 1992) and a species (*Polygonum hydropiper*) typical for very wet soils (Ellenberg et al. 1992), a plant community which can be interpreted to indicate a disturbance in a stream habitat.

The community of larval dragonflies of Site 1 was characterized by the dominance of *Ischnura elegans* which accounted for more than 64% of the total number of exuviae, one of the most euryoecious European species (Heidemann & Seidenbusch 1993; Sternberg & Buchwald 1999). Rehfeldt (1983) suggested that the dominance of *I. elegans* might indicate running water systems which are heavily contaminated. Heidemann and Seidenbusch (1993) extended this observation to a variety of different

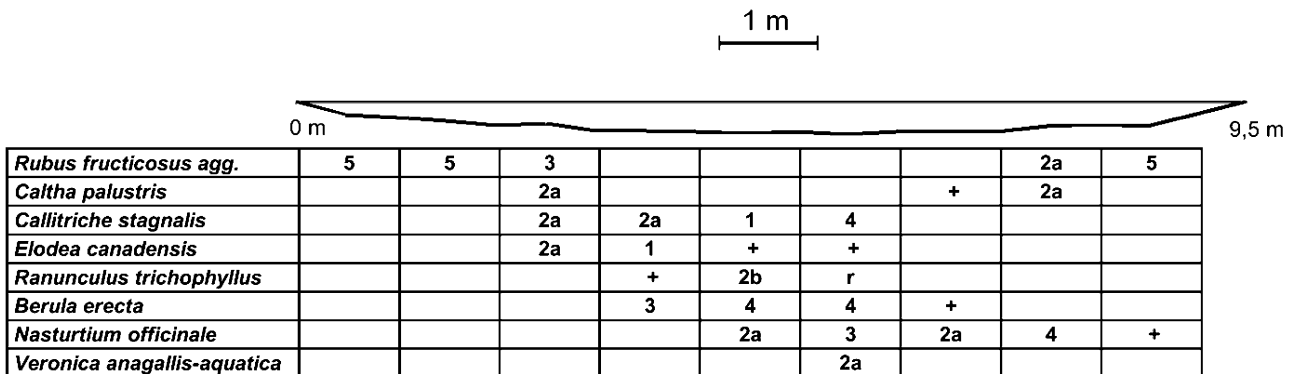


Figure 8. Site 2: morphology and vegetation (cover according to Reichelt & Wilmanns (1973); r: 1 individual, +: 2–5 individuals (<5% cover), 1: 6–50 individuals (<5% cover), 2m: > 50 individuals (<5% cover), 2a: cover 5–15%, 2b: cover 16–25%, 3: cover 26–50%, 4: cover 51–75%, 5: cover 76–100%).

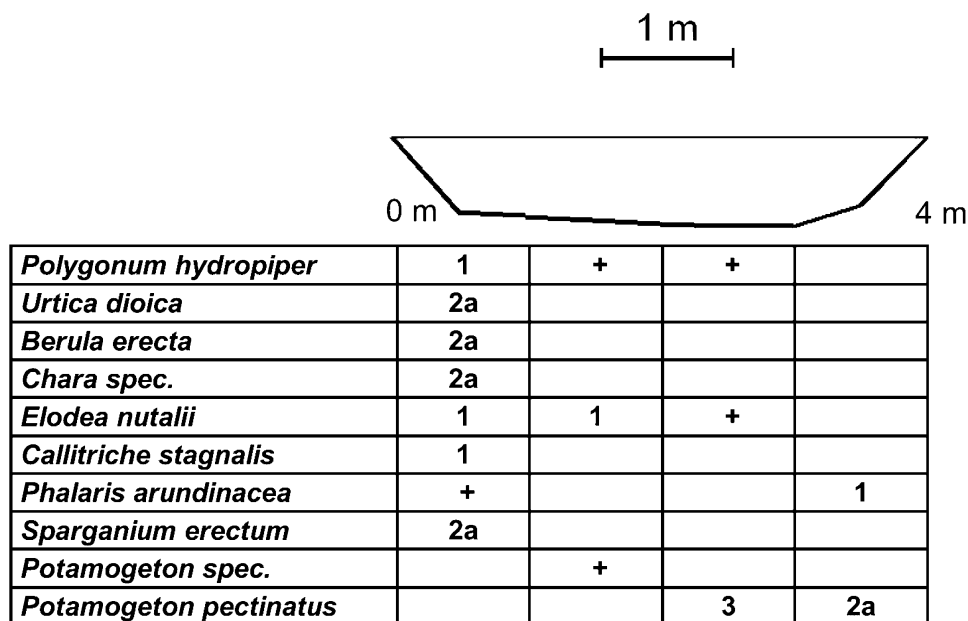


Figure 9. Site 3: morphology and vegetation (cover according to Reichelt & Wilmanns (1973); r: 1 individual, +: 2–5 individuals (<5% cover), 1: 6–50 individuals (<5% cover), 2m: > 50 individuals (<5% cover), 2a: cover 5–15%, 2b: cover 16–25%, 3: cover 26–50%, 4: cover 51–75%, 5: cover 76–100%).

abiotic factors (contamination, salinity, oxygen saturation) and wrote that *I. elegans* can survive under conditions which no other (European) dragonfly species can tolerate.

Libellula fulva Müller, 1764, in river systems a species typical for downstream mud habitats (Schridde & Suhling 1994), was more common at Site 1 in comparison with the other two sites. Exuviae of *Aeshna mixta* (Latreille, 1805) and *Anax imperator* Leach, 1815 were exclusively found here. All these species are typical for stagnant water and waters with a weak current. Species typically found in stagnant waters made up for approximately 66% of all exuviae at Site 1. The larvae of *Platycnemis pennipes* are generally found in running or in slow-moving waters and occur regularly also in habitats with only slight water movements (e.g. Castella 1987). *P. pennipes* was the only species which occurred in all three sites in similar frequencies. Three exuviae of *Calopteryx splendens* (Harris, 1782) (accounting for approximately 1% of all exuviae), another species generally considered typical for

running waters, were collected. However, larvae of *C. splendens* are known to occasionally occur in stagnant water (Carchini 1983; Hofmann & Mason 2005b). Larvae of species which exclusively occur in running water such as *Gomphus vulgatissimus* (Linnaeus, 1758) or *Onychogomphus forcipatus* (Linnaeus, 1758) were not found in Site 1, as hypothesized. In contrast, Ferreras Romero et al. (1995) reported that *O. forcipatus* was found to complete its lifecycle in watercourses that are reduced to a series of isolated ponds from July to October. However, they state that this subject requires further investigation.

Site 2 retained a continuous water flow throughout the entire period observed (Figure 6) and observations from recent years suggest that this part of the Rio Begotta is always characterized by flowing water. This stretch of the Rio Begotta showed the lowest fluctuations in water level, the difference between highest and lowest levels being less than 30 cm. The dominant plant species in the river bed were *Berula erecta* and *Nasturtium officinale*, which indicate spring-water (Corbetta 1969; Albergoni et al. 1977). *Ranunculus trichophyllus*, a plant species which indicates permanent streams (Ellenberg et al. 1992), was present and species which indicate disturbances were absent. At Site 2 typical rheophilous larvae accounted for more than 50% of the exuviae and species typical of stagnant water were absent.

Table VI. Average water temperatures (°C) of the three sites investigated (for details see text).

	Winter	Spring	Summer	Autumn
Site 1	4.0	18.1	21.5	17.5
Site 2	7.7	17.0	20.4	17.9
Site 3	8.7	18.5	22.4	19.0

Site 3 provided a water depth of more than 30 cm throughout the entire period. With 0.8 m s^{-1} the water flowed fastest at Site 3. Another factor to consider is the pronounced fluctuations of the water level of up to 25 cm in 5 days. These occurred between mid April and early October and were presumably caused by water management for agricultural purposes. *Potamogeton pectinatus*, a plant which indicates permanent waters (Ellenberg et al. 1992), was the dominant species in the stream. At Site 3 a total of 10 species were collected as exuviae, with rheophilous species totalling over 64%. Species typical for stagnant water made up for 8% of the community.

The adult communities

The communities of adult dragonflies (Table III) observed did show some site-specific differences. However, their compositions were relatively similar, with similarity indexes always exceeding 0.97, as hypothesized. The dominant species were *Calopteryx splendens*, *Orthetrum coerulescens* and *Platycnemis pennipes* at all three sites. A pronounced difference was the abundance of *I. elegans*, which was 9% at Site 1 and only 0.3% at Site 2. This difference can be explained by the fact that at Site 1 *I. elegans* made up for 64% of all exuviae. In contrast, at Site 2 no exuviae of this species were found. It is known that only a small number of *I. elegans* do disperse and that these dispersing animals do not normally fly far (Conrad et al. 1999). The difference in the adult frequencies for *C. splendens* can probably be explained by the fact that Site 2 presents the best habitat for this species, with a well developed underwater vegetation and many lateral channels with a modest flow rate, and this site was probably selected by the species. The reason for the observed difference in the adult frequencies for *C. coerulescens* are less obvious.

Comparison of larval and adult communities

Comparing the similarities of larval and adult communities for the three sites revealed that the species compositions differed more at Site 1 than in Sites 2 and 3, as hypothesized. For Site 1 the main difference between larval and adult communities was that adults of *Calopteryx splendens* and *Orthetrum coerulescens* frequented Site 1 in large numbers even though exuviae of these species were absent or very rare (3 exuviae of *C. splendens*). In contrast, *C. splendens* was repeatedly observed at Site 1 while ovipositing and *O. coerulescens* was repeatedly seen in copula. Another rheophilous species which was

present at Site 1 as adults and of which no exuviae were found was *Onychogomphus forcipatus*. The likely cause for the absence of exuviae of these species is the hydrological situation, the little water present from the end of March to the end of April being stagnant. Also Rossi et al. (2003) showed that larvae of *O. forcipatus* do not survive drying out during autumn–winter. The temporary interruption of the water flowing might not allow for the survival of the larvae of these rheophilous species. However, during the flight period of the adult dragonflies (mid May to October), the Rio Begotta at Site 1 presented a water depth of 20–30 cm and a swift flow. The fact that the vegetation in Site 1 was not typical for a small stream does not seem to have affected the identification of a suitable habitat by the adult dragonflies.

It is known that the flow of water is an important factor for *C. splendens*. For the larvae, $3\text{--}30 \text{ cm s}^{-1}$ seems to be the optimal range and adult males with territories comprising a section of a stream with water flowing at 15 cm s^{-1} are most successful reproductively (Sternberg & Buchwald 1999). Male *C. splendens* have a complex courtship behaviour which involves, amongst other behaviour, the male displaying the oviposition site by alighting on the water surface near it – the so-called “dive display”. This shows the female the quality as well as the location of the site because the courting male often allows itself to be carried several centimetres downstream by the water current (Corbet 1999). Thus, in *C. splendens* flowing water seems to be a “proximate factor”, indicating a suitable habitat. This is likely to be true also for *Orthetrum coerulescens* and *Ophiogomphus forcipatus*. If this is correct, the flow of water during the reproductive period at Site 1 represented an “ecological trap” (Schlaepfer et al. 2002; Robertson & Hutto 2006) for rheophilous dragonflies because environmental change, created by human intervention, uncouples the cues that individuals use to assess habitat quality from the true quality of the environment (Robertson & Hutto 2006). An ecological trap was shown to exist as the evidence required according to Robertson and Hutto (2006) was presented:

1. Individuals exhibited an equal preference for the altered environment as the communities of adults were very similar for all three sites.
2. The fitness of individuals varied by habitats/sites.
3. The rheophilous species had a lower fitness in the altered environment.

The rheophilous dragonfly species recognized a suitable habitat by means of proximate factors when

in reality habitat conditions were not suitable for the larvae. However, even in habitats which are not heavily modified, the selected sites for oviposition are not always optimal for the development of the larvae (Wildermuth 1994).

In contrast, at Sites 2 and 3, where the hydrological situation was more natural, the “proximate factor” correctly indicated a suitable habitat for rheophilous species.

Of the 21 species observed as adults, 16 contributed on average to less than 1% of the community. Amongst these is also *Onychogomphus forcipatus*, the exuviae of which made up for 16% and 22% of larval communities at Sites 2 and 3, respectively, while adults comprised only 0.5–0.7% of all Odonata observed. These discrepancies show that comparison of larval populations (by means of exuviae) and adult abundance has to be interpreted with caution and that one has to consider adult behaviour as well. The scarcity of adult Gomphidae near the water is a well-known fact. For example, Breuer (1987) stated for *Gomphus vulgatissimus* that exuviae were found in large numbers whereas adults were rarely observed near the water. Corbet (1999) wrote “especially, it seems, among gomphids, many exuviae but no mature adults may be found at a site”. Therefore the high numbers of exuviae of *O. forcipatus* found at Sites 2 and 3 show that these aquatic habitats were suitable for larval development. In this species, differences in the presence of larvae and adults do not indicate problems of habitat suitability.

Emergence patterns of selected species

The only species which emerged in sufficient numbers at all three sites to allow a comparison was *P. pennipes*, which in Site 2 reached the median 33 days and 30 days later than in Sites 1 and 3, respectively. Site 2 was on average more than 1°C colder during spring and summer. This result confirms the observation that lower average water temperatures result in a later emergence (e.g. Westermann 2002).

Conclusions

One result of this study is the fact that at Site 1, which had flowing water only from late spring to late autumn, in the larval community the rheophilous species were almost entirely absent. In contrast, the majority of adult Odonata observed is typical for flowing water habitats. This result has important implications for dragonfly surveys as it shows that the analyses of the adult fauna can be misleading (see also Amico et al. 2004; Hofmann & Mason

2005a). Moreover, the results suggest that stream systems such as Site 1, which are not uncommon in the Po Valley (e.g. Rossi et al. 2003), may function as ecological traps for rheophilous species, some of which are rare. The adult dragonflies recognize suitable habitats by means of “proximate factors” when in reality they do not allow the larvae of these specialized species to survive (see also Hofmann & Mason 2005a), because the sites fall dry for some time during the period from autumn to early spring. Such ecological traps can contribute to population extinction, particularly at low population densities, as they constitute “habitat sinks” (Schlaepfer et al. 2002).

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